Dynamic Intersection Risk Analysis for Proactive ADAS

A Project Report Submitted by

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in partial fulfillment of the requirements for the award of the degree of

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Department of Civil and Infrastructure Engineering

Indian Institute of Technology Jodhpur

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Declaration

I hereby declare that the work presented in this Project Report titled "Dynamic Intersection Risk Analysis for Proactive ADAS" submitted to the Indian Institute of Technology Jodhpur in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology, is a Bonafide record of the research work carried out under the supervision of Dr.Ranju Mohan. The contents of this project report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Prathuri Vinoothna

B22CI030

Certificate

This is to certify that the Project Report titled "Dynamic Intersection Risk Analysis for Proactive ADAS", submitted by Prathuri Vinoothna(B22CI030) to the Indian Institute of Technology Jodhpur for the award of the degree of Bachelor of Technology, is a Bonafide record of the research work done by her under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

Dr. Ranju Mohan

Abstract

Project Crossroads introduces an innovative way of analyzing the risk of intersections proactively using bird's eye view (BEV) video as part of next generation Advanced Driver Assistance System (ADAS). Project Crossroads converts raw video from an unstructured source to a structured pipeline which includes perception, kinematics analysis, safety surrogate metrics and predictive modelization. Infrastructure and road user's motion is initially annotated so that a customized YOLO detector and homography-based transformation can be used to create realistic vehicle trajectories. Using ByteTrack, the motion history of all moving objects can then be computed to determine velocity, acceleration and other key safety metrics, including Post Encroachment Time (PET), Time To Collision (TTC), gap acceptance, and stop line compliance. Agent – infrastructure interaction is modelled by a Heterogeneous Graph Transformer (HGT) that will predict the short-term risk score of each driver and classify their behaviour. Project Crossroads also has the capability to provide real-time visualisations of behaviour, analytical behaviour data, and counterfactual ADAS intervention studies. Ultimately, this method of evaluating the risk associated with intersections in India is scalable and will provide tangible insight to the future development of both intelligent transportation System.

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1. Introduction

Intersections are busy, unpredictable places where a lot can go wrong in just a few seconds. In India, with so many types of vehicles and different driving habits, it becomes even tougher to understand what actually creates risky situations on the road. That's where Project Crossroads comes in. Using simple bird's-eye-view video, it observes how vehicles move, how close they come to each other, and when a moment might turn dangerous. The system tracks each vehicle, studies its behavior, and looks for patterns that could lead to near-misses. The idea is straightforward: understand what's really happening at intersections so we can make them safer. Project Crossroads turns everyday traffic movement into clear insights that can support smarter, more effective road-safety solutions.

2. Background

Intersections are usually where the road feels the most hectic. In India, it's even more intense — cars squeezing through gaps, bikes weaving around everything, autos stopping anywhere, buses pushing through, and pedestrians trying to cross however they can. With so many people moving in their own ways, it's tough to tell what actually causes those sudden risky moments or near-misses. And traditional methods like reports or someone watching from the side of the road simply can't capture everything happening second by second.

But with better video tools and AI, we finally have a clearer way to understand all this chaos.A simple bird's-eye-view video can show the entire intersection at once, revealing how vehicles approach, how they react to each other, and where things start to get unsafe. From above, patterns and mistakes that usually go unnoticed suddenly become obvious.That's exactly what Project Crossroads focuses on. It takes regular overhead videos and turns them into real insights about how people actually behave at intersections. By tracking every vehicle and studying how they move around one another, the project tries to figure out what triggers those dangerous situations. The goal is simple: understand the reality on the road so we can make intersections safer and help build smarter driving systems in the future.

3. Literature Survey

Researchers have been trying to understand traffic behavior at intersections for many years.Earlier work mostly depended on manual observation or basic camera footage.While these methods helped identify general traffic flow and rule-breaking, they often failed to catch the small but important moments—like a sudden brake, a risky turn, or two vehicles narrowly avoiding each other. These near-miss events are crucial because they reveal danger long before an actual accident happens.As technology improved, studies began using computer vision to automatically detect and track vehicles. Models like YOLO became widely used because they can quickly recognize different objects in a video, making them ideal for busy roads. Tracking methods also improved over time. Algorithms such as byte track made it possible to follow each vehicle smoothly, even when there were many vehicles close together.Another major area of research focuses on measuring risk without waiting for accidents to occur. This led to the use of safety indicators like Time-to-Collision (TTC) and Post-Encroachment Time (PET). Many studies show that these measurements provide a clear picture of how dangerous an interaction is. Since actual crashes are rare and unpredictable, these "near-miss" indicators have become an important tool in traffic safety research. More recent work has started exploring how vehicles influence each other. This is where graph-based and transformer models come in. These models treat each vehicle as a separate element and study how they interact—like who is yielding, who is cutting across, or who is stopping suddenly. This gives a deeper understanding of behavior, not just movement.

Overall, the literature shows a clear move from simple counting and observing to deeper, AIpowered analysis. Detection, tracking, near-miss indicators, and interaction modeling have all been widely studied. These foundations strongly support Project Crossroads, which brings these ideas together to understand intersection behavior in a practical, real-world way.

4.Problem Definition and Objective

The aim of this project is to understand what actually happens at busy road intersections and use that knowledge to improve safety. Instead of just detecting vehicles in a video, the idea is to look at how they move, how they interact with each other, and what leads to those sudden risky moments that often go unnoticed.

By analysing bird's-eye-view footage, the system tries to follow each road user and observe their behaviour in a realistic way—how fast they go, how they respond to other vehicles, and how close they come to having a near-miss. These small details help us see the road the way a human would, but with the consistency and precision of a computer.

In simple terms, the goal is to build a system that can spot danger a few seconds before it happens and turn that insight into something useful for ADAS and traffic safety. If successful, it can help drivers make safer decisions and give traffic planners a clearer understanding of what's really happening on the roads.

5.Phase 1: Foundation – Scene Definition and Static Analysis

5.1 Infrastructure & Road Mapping

Before jumping into any modelling, I had to get a proper feel of the intersection. I was working with. So in Phase 1, my main focus was to build the "base layer" of the project — basically understanding how the road is laid out and how everything fits inside the frame.I started by picking a few clear frames from the bird's-eye-view video. These frames became my reference map. Using an annotation tool, I manually traced the important static parts of the intersection. These elements don't change across frames, so I saved all of them into a JSON configuration file. This file basically became my static road map. It helped me understand where each vehicle is on the road not just where it appears in the video. Having this fixed layout made later stages much easier and more reliable.

5.2 Dynamic Object Annotation

Once the static parts were ready, I moved on to annotating the things that *do* move. I used CVAT to label around 150–200 frames so I could cover different traffic conditions like peak hours, slow movement, and mixed-vehicle flow.

For each frame, I drew bounding boxes around:

- Cars
- Bikes and motorcycles
- Auto-rickshaws
- Buses

- Trucks
- Pedestrians

I chose these classes because they represent what you normally find on Indian roads. The idea was to build a clean and accurate dataset that could later be used to train a custom YOLO model. By labeling so many frames, the model gets a good understanding of how each object looks from a top-down view.By completing both the static road map and the dynamic annotations, Phase 1 gave me a solid foundation to build the rest of the system on. build a clean and accurate dataset that could later be used to train a custom YOLO model. By labeling so many frames, the model gets a good understanding of how each object looks from a top-down view.By completing both the static road map and the dynamic annotations, Phase 1 gave me a solid foundation to build the rest of the system on.

6. Phase 2: Perception – Real-Time Scene Understanding

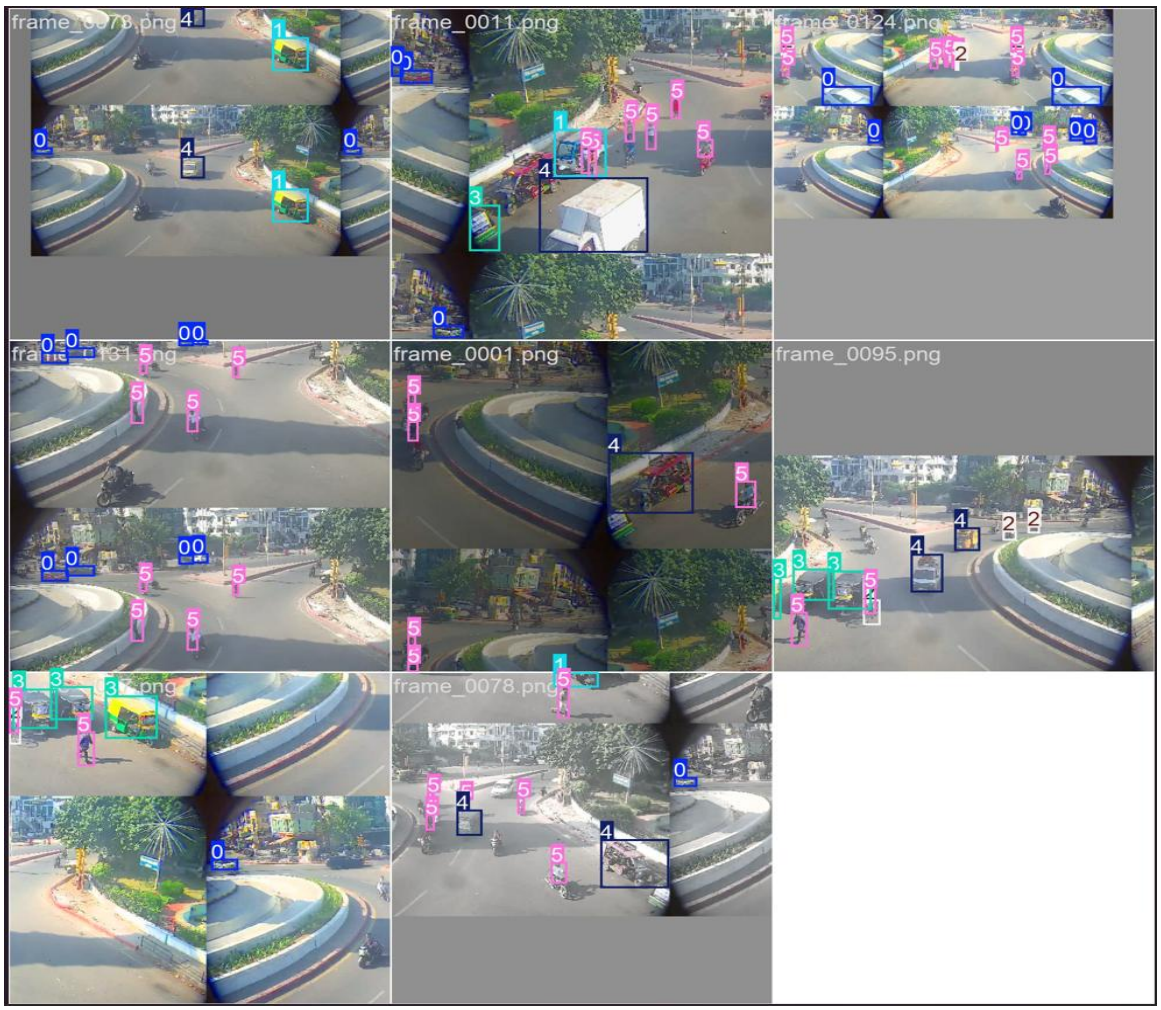
6.1 Custom YOLOv11 Training

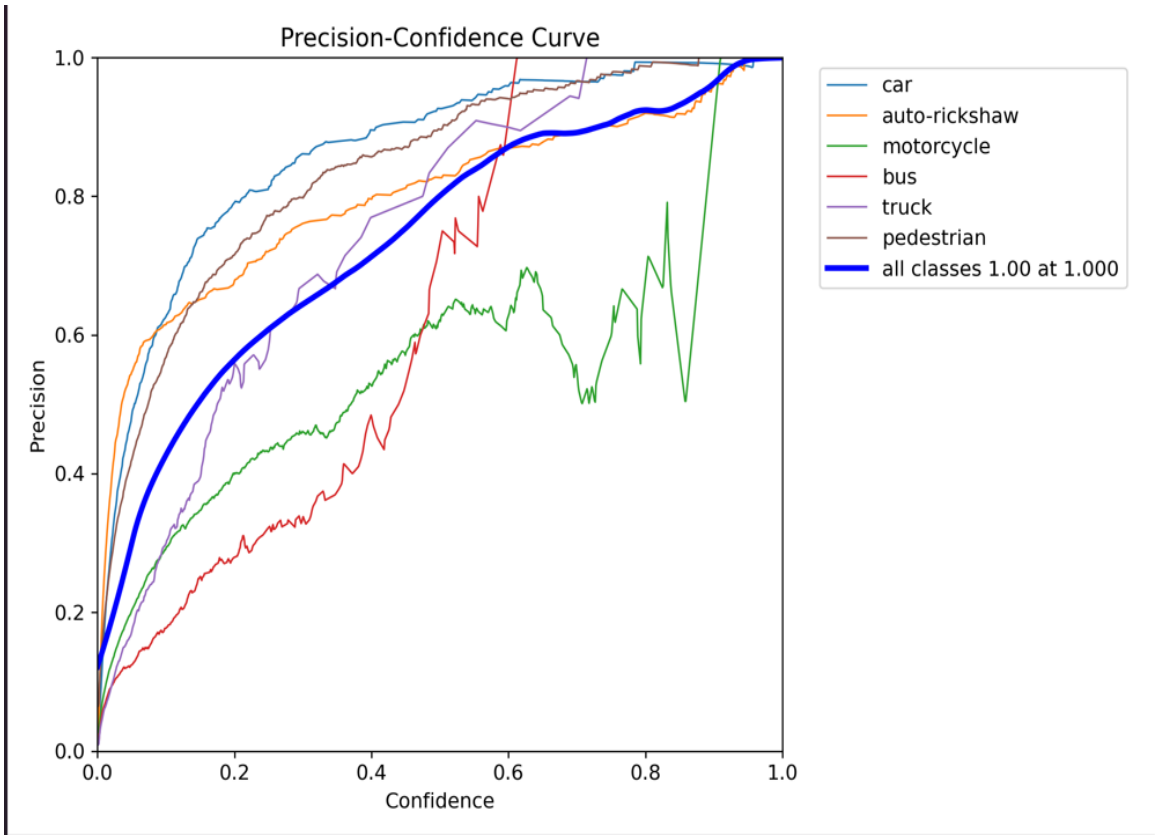
Using the annotated dataset from Phase 1, I trained a custom YOLOv11 model. Since the camera angle is fixed and the view is from the top, I needed the model to be specifically tuned for this setup — general YOLO models don't work well for top-down traffic.

Steps I followed:

1. Exported the CVAT labels to YOLO format.
2. Created train/val/test splits.
3. Trained YOLOv11 using the recommended hyperparameters.
4. Fine-tuned the model until both precision and recall improved.

The final model was able to detect cars, bikes, autos, buses, trucks, and pedestrians with good accuracy. This detection output became the base input for all later calculations.





6.2 BEV Coordinate Transformation

Once the model could detect objects, the next issue was:

detections are in pixels, but real traffic analysis needs meters.

To solve this, I used a **homography transformation**. I selected four or more known reference points on the road (like lane corners or crosswalk edges). Using these points, I calculated a matrix that converts image coordinates (u, v) → ground coordinates (X, Y).

This transformation allowed me to:

- measure distances in meters,
- calculate speeds,
- understand vehicle interactions in physical space.

This step is really important because all upcoming risk metrics (like PET and TTC) depend on real-world values.

6.3 Multi-Object Tracking

Object detection alone is not enough — each frame is separate. For traffic analysis, I needed

to follow each vehicle smoothly over time.

I implemented **ByteTrack**, a tracking algorithm that assigns a unique ID to every detected vehicle and keeps it consistent across frames.

This step allowed me to:

- create continuous trajectories,
- study movement patterns,
- link positions across time,
- form the base dataset for velocity and acceleration calculations.

By the end of Phase 2, I had:

- accurate detections,
- real-world coordinates,
- and complete vehicle trajectories.

This sets up Phase 3, where I convert these trajectories into measurable safety and behavior metrics.

7. Phase 3: Quantification – From Traectories to Risk Metrics

Phase 3 it is where the project starts converting all the tracked vehicle data into meaningful numbers. Instead of just knowing where each vehicle is, I now calculate how it is moving (speed, acceleration) and whether it is involved in any risky situation using standard Safety Surrogate Measures (SSMs). These metrics help identify near-misses and driver behavior patterns without needing real crashes to occur.

7.1 Kinematic State Estimation

Kinematic state estimation is the process of extracting the basic movement characteristics of each vehicle from its trajectory. Since I already had the real-world (X, Y) positions of every vehicle in each frame (after homography and tracking), I used this information to compute important variables like distance, velocity, and acceleration.

1. Distance Between Frames

Distance is used to understand how much a vehicle has moved and to also calculate spacing between vehicles.

$$D = \sqrt{(X_t - X_{t-1})^2 + (Y_t - Y_{t-1})^2} \tag{1}$$

This formula gives the Euclidean distance between two consecutive positions. It forms the basis for computing speed and other metrics.

2. Velocity (Speed) Calculation

Velocity shows how fast the vehicle is moving between frames. I calculated horizontal and vertical components first:

$$V_x = \frac{X_t - X_{t-1}}{\Delta t}, V_y = \frac{Y_t - Y_{t-1}}{\Delta t} \tag{2}$$

Then the total speed:

$$V = \sqrt{V_x^2 + V_y^2} \tag{3}$$

This helps identify if a vehicle is speeding, slowing down, cruising steadily, or accelerating.

3. Acceleration Calculation

Acceleration represents how quickly the velocity changes between frames:

$$A_x = \frac{V_{x,t} - V_{x,t-1}}{\Delta t}, A_y = \frac{V_{y,t} - V_{y,t-1}}{\Delta t} \tag{4}$$

Total acceleration:

$$A = \sqrt{A_x^2 + A_y^2} \tag{5}$$

High acceleration or sudden negative acceleration (braking) often indicates risky interactions, unexpected stops, or harsh maneuvers.

4. Stopping Distance

Stopping distance is used to understand if a vehicle could stop safely under emergency conditions:

$$d_{\text{stop}} = \frac{V^2}{2a} \tag{6}$$

This helps identify whether the vehicle had enough distance to stop before the stop line or before hitting a slower vehicle.

7.2 SSM & Driver Behavior Metrics

Safety Surrogate Measures (SSMs) are mathematical indicators used to quantify the risk level of interactions between vehicles. These metrics help detect near-misses and dangerous behavior without needing actual crashes, making them extremely important for traffic safety research.

In this phase, I computed four major SSMs: PET, TTC, Gap Acceptance, and Stop Line Compliance.

• Post-Encroachment Time (PET)

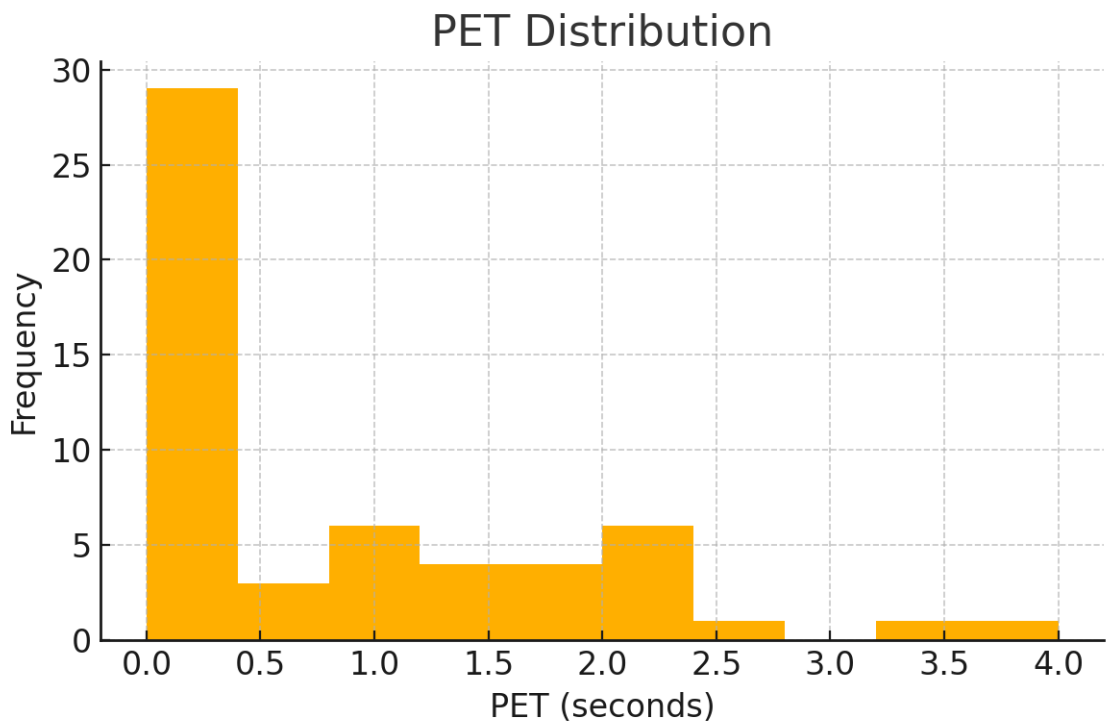
PET is one of the most reliable indicators for intersection safety. It measures how closely two vehicles pass through the same conflict point.

$$PET = T_{\text{vehicle B arrives}} - T_{\text{vehicle A leaves}} \tag{7}$$

- If PET is **small**, the vehicles passed through the same location almost at the same time.
- PET below **1.5 seconds** is generally considered a **near-miss**.

Purpose:

PET helps identify dangerous crossing, merging, and turning situations, especially at uncontrolled or busy intersections.



• Time-to-Collision (TTC)

TTC estimates how much time is left before two vehicles would collide **if they maintain their current speed and direction**.

For rear-end scenarios:

$$TTC = \frac{D}{V_{\text{following}} - V_{\text{leading}}} \tag{8}$$

Where:

- **D** is the distance between the vehicles
- If the following vehicle is not faster → TTC = ∞ (no risk)

Interpretation:

- TTC < **3 seconds** → high-risk interaction
- Rapidly decreasing TTC → potential crash or harsh braking event

TTC is especially useful for identifying tailgating and sudden braking situations.

• Gap Acceptance

Gap Acceptance measures how much time gap a driver chooses before crossing or merging into traffic.

$$\text{Gap} = \frac{D_{\text{oncoming}}}{V_{\text{oncoming}}} \tag{9}$$

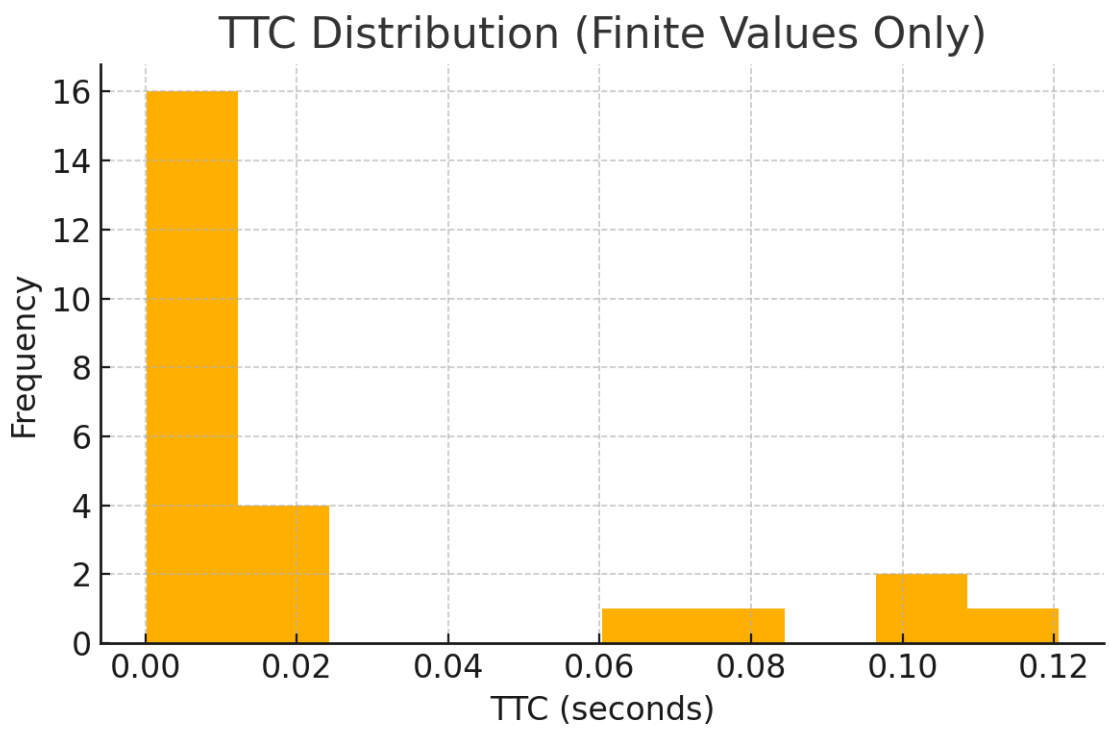
Where:

- **D** = distance of the oncoming vehicle
- **V** = speed of the oncoming vehicle

Interpretation:

- **Small accepted gap** → aggressive driver
- **Large gap** → defensive/cautious driver

This metric helps understand turning behavior, especially right-turns and pedestrian crossing decisions.



• Stop Line Compliance

This metric checks whether drivers follow the rule of stopping before the stop line.

Distance from stop line:

$$D_{stop} = \sqrt{(X_s - X_l)^2 + (Y_s - Y_l)^2} \tag{10}$$

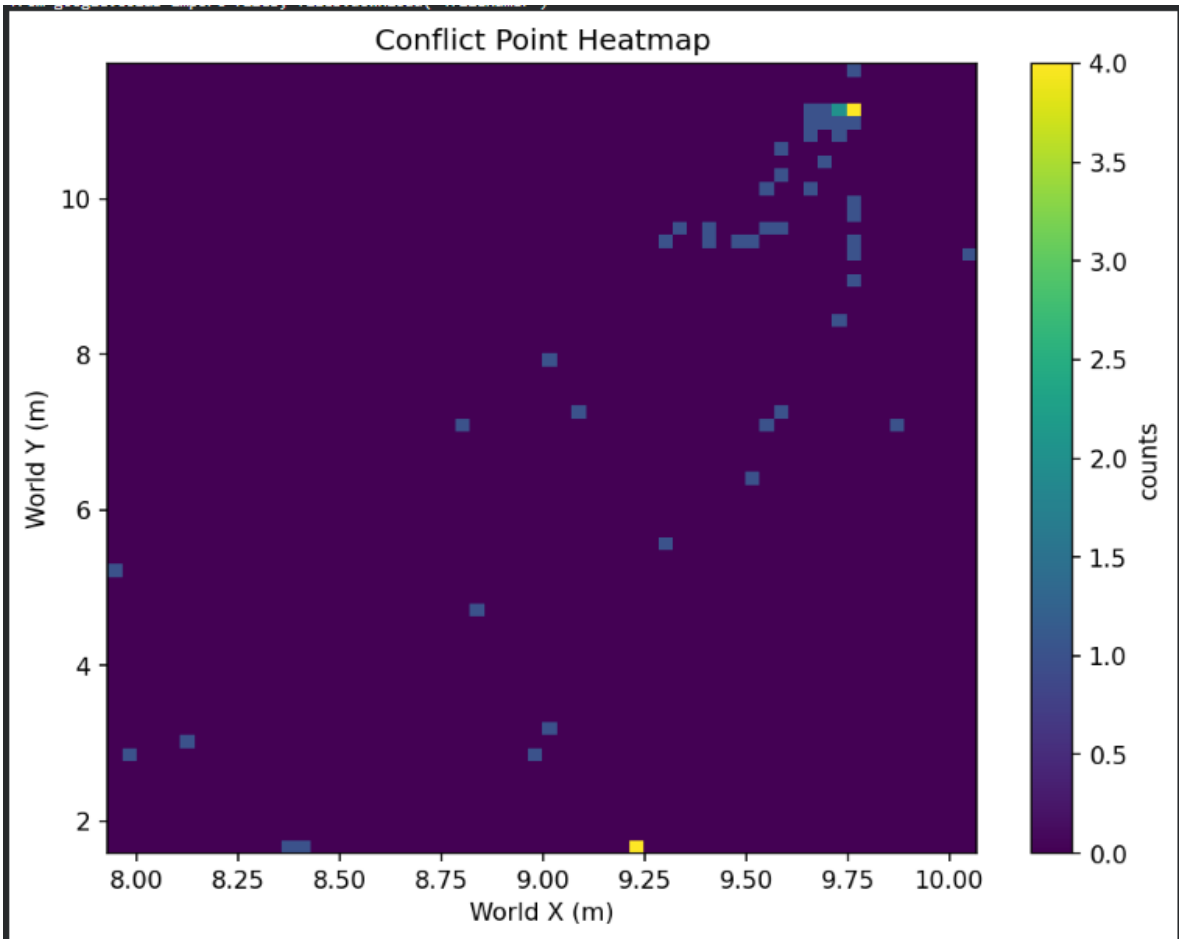
Where:

- (X_s, Y_s)= point where vehicle actually stopped
- (X_l, Y_l)= stop line location

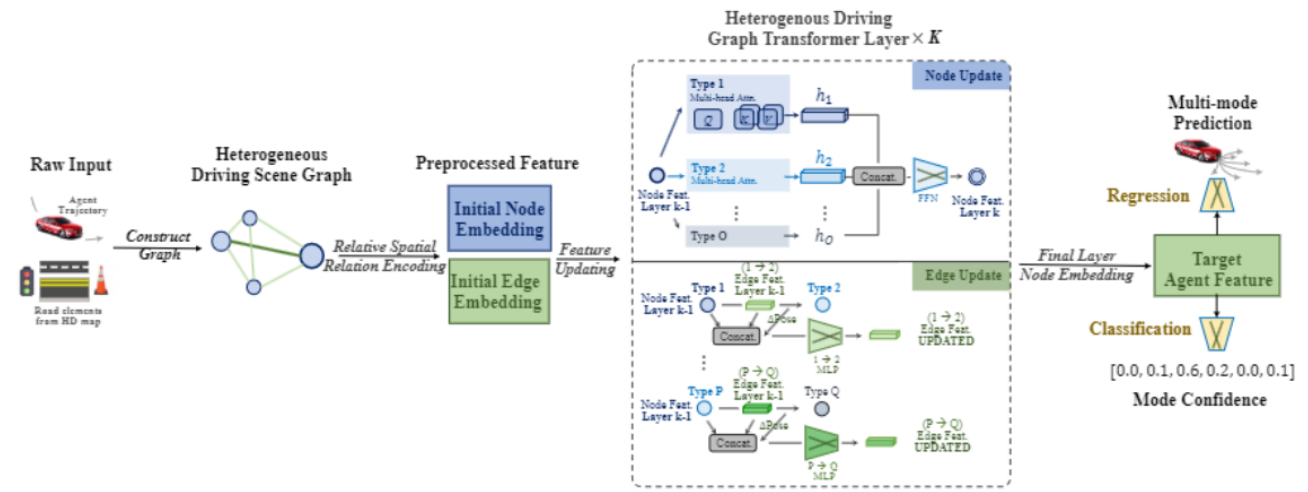
Interpretation:

- Negative or very small D → vehicle crossed the stop line (violation)
- Large D → cautious but possibly inefficient stopping behavior

Stop line compliance shows discipline and rule-following at signals or yield points.



8. Phase 4: Advanced Modeling – Heterogeneous Graph Transformer



Phase 4 focuses on converting the processed traffic data into a structured graph format and then using a Heterogeneous Graph Transformer (HGT) model to understand how all the agents in the intersection interact with one another. The idea is to move beyond analyzing vehicles individually

and instead study the entire scene as a connected system. This allows the model to capture how cars, bikes, pedestrians, and road infrastructure jointly influence risk and driving behavior.

8.1 Heterogeneous Graph Construction

To represent each frame of the traffic video in a meaningful way, a heterogeneous graph is constructed using two main components: nodes and edges. The node set includes both vehicle nodes and infrastructure nodes. Vehicle nodes represent every tracked agent (car, auto-rickshaw, motorcycle, bus, truck, pedestrian) and store features such as position, velocity, acceleration, heading direction, and safety metrics like TTC, PET, and DRAC. Infrastructure nodes include stoplines, lane boundaries, pedestrian crossings, and conflict points, which provide the structural context of the road. The edge set defines the relationships between these nodes. Vehicle-to-vehicle edges capture relative distance, speed difference, and potential conflicts. Vehicle-to-infrastructure edges represent how close an agent is to elements like stoplines or lane boundaries. Temporal edges link the same vehicle across consecutive frames, helping the model understand motion over time. Before feeding the graph into the model, each node and edge is converted into an embedding vector so that all objects can be processed numerically.

8.2 Predictive Model: Heterogeneous Graph Transformer (HGT)

Once the graph is created for every frame, it is passed through the Heterogeneous Graph Transformer. The reason for using an HGT instead of a normal graph neural network is that traffic scenes involve different types of agents and relationships. A car interacting with a motorcycle is not the same as a car interacting with a stopline, so the HGT uses separate attention mechanisms for each type of node and edge. Each HGT layer updates both node features and edge features by looking at the surrounding context, attention weights, and temporal information. Through multiple layers, the model learns patterns in traffic behavior, such as overtaking, yielding, merging, or entering a conflict zone. The final output of the HGT consists of two predictions for every vehicle: a risk score between 0 and 1 indicating the likelihood of a near-miss in the next few seconds, and a driver behaviour class (Normal, Aggressive, or Hesitant), based on the vehicle's recent motion patterns and safety metrics.

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    print ("Model saved to phase4_hgt_model.pt")
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Epoch 03: Train loss 0.624  Val loss 0.552  Val loss:0.59
Epoch 04: Train loss 0.672  Val loss 0.552  Val biaac:667
Epoch 05: Train loss 0.456  Val loss 0.592  Val betac:663
Epoch 06: Train loss 0.452  Val loss 0.558  Val bvacc:663
Epoch 07: Train loss 0.493  Val loss 0.552  Val altac:663
Epoch 08: Train loss 0.491  Val loss 0.558  Val al03 :676
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phase_hgt.py

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Epoch 02 | Train loss: 0.8429 | Val loss: 0.5792 | Val beh acc. 0.
Epoch 03 | Train loss: 0.6724 | Val loss: 0.7449 | Val beh acc. 0.
Epoch 04 | Train loss: 0.542| | Val loss: 0.5436 | Val beh acc. 0.
Epoch 05 | Train loss: 0.4323 | Val loss: 0.5622 | Val beh acc. 0.
Epoch 08 | Train loss: 0.3556 | Val loss: 0.5409 | Val beh acc. 0.
Epoch 07 | Train loss: 0.2064 | Val loss: 0.3427 | Val beh acc. 0.
Epoch 08 | Train loss: 0.3562 | Val loss: 0.4938 | Val beh acc. 0.
Epoch 09 | Train loss: 0.2945 | Val loss: 0.4957 | Val beh acc. 0.
Epoch 10 | Train loss: 0.1770 | Val loss: 0.4762 | Val beh acc. 0.
Epoch 11 | Train loss: 0.1614 | Val loss: 0.4763 | Val beh acc. 0.
Model saved to phase4_hgt_model.pt

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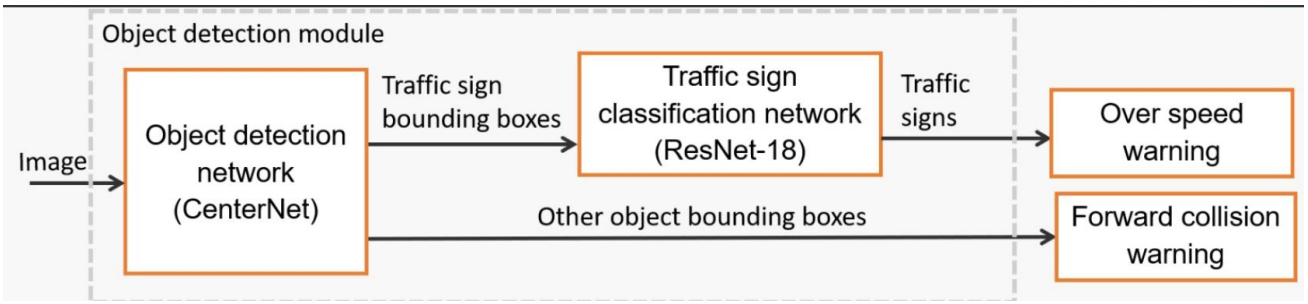
9. Phase 5: Implementation – ADAS & Safety Analysis



When I got to Phase 5, the model was already performing well in detection, tracking, and risk estimation. However, figures by themselves do not reveal the complete picture. This phase focused on translating all the system's predictions into a format that is comprehensible and usable for people.

The main objective was to align the model with real-world safety applications—particularly regarding ADAS—and demonstrate how these predictions can aid in explaining risky actions or potentially avert accidents. This stage marks the transition of the project from merely being “a computer vision pipeline” to transforming into a genuine tool for traffic understanding

9.1 Visualization in Real-Time

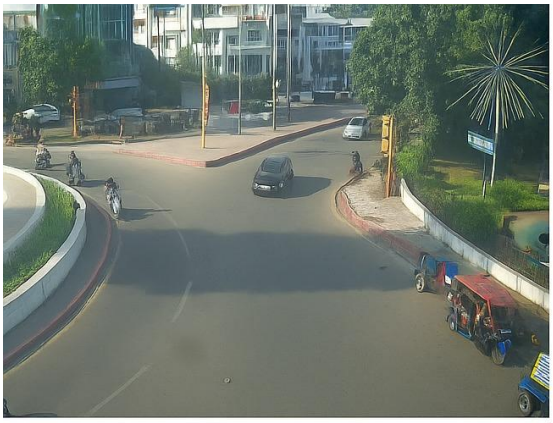
In this phase, I initially created a real-time visualization that displays the model's predictions directly on the original video. This made the entire situation seem far more “vibrant.” Rather than merely viewing TTC values in a table, you can observe which vehicle poses a risk at any given time, as the bounding box alters its color—green if it’s safe, yellow if something ambiguous is occurring, and red when it enters a near-miss area. I added small tags indicating the type of maneuver the vehicle appears to be performing, such as a quick stop or a sharp turn.

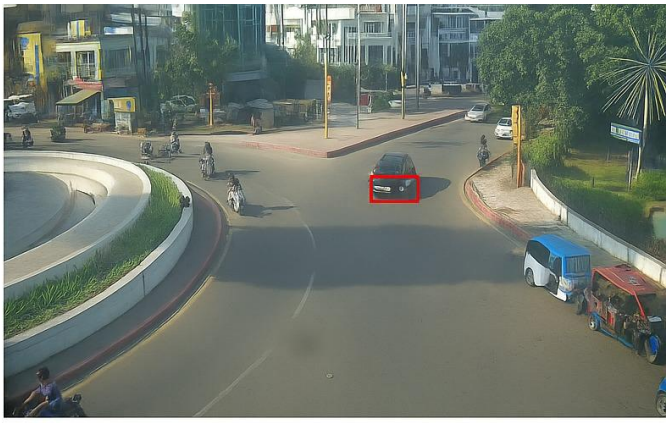
As you view the video featuring these overlays, patterns begin to emerge spontaneously. You observe specific corners frequently glowing red, or cars that regularly act erratically. This live perspective simplified the process of verifying whether the model was functioning as intended, and it also turned out to be one of the most effective methods for showcasing the complete system to others

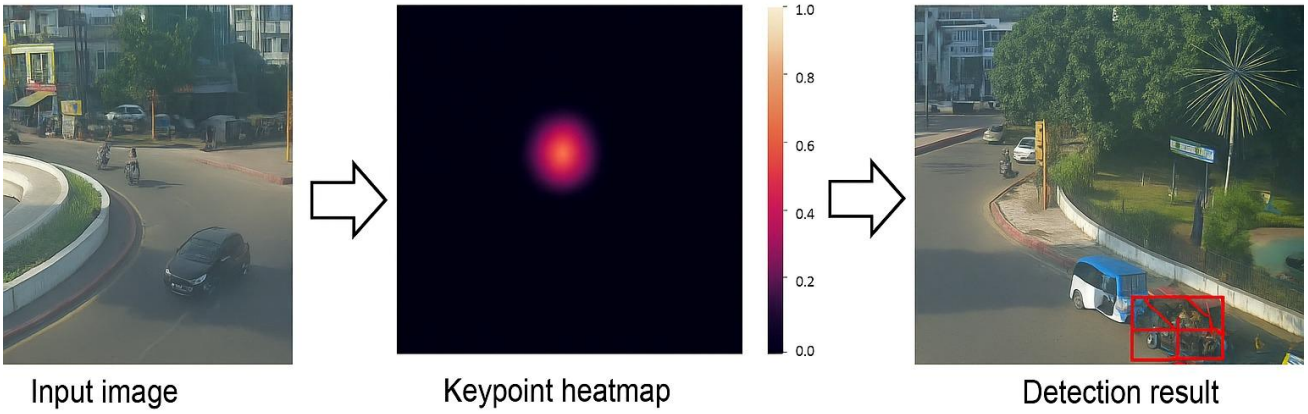
9.2 Aggregate Driver Behavior Analysis

Following that, I shifted to a significantly broader perspective by executing the model on extended sequences of video. This allowed me to grasp not only individual moments but also the general behavior of various vehicles at the intersection. As time passed, trends emerged distinctly. Two-wheelers, for instance, were the most daring—they would slip through openings and speed up rapidly. Vehicles were more fluid but still triggered abrupt lane shifts occasionally. Vehicles were the most uncertain regarding stopping, particularly at merging areas.

Using this processed data, I was able to create graphs and heatmaps illustrating the locations of frequent near-misses, identifying which maneuvers posed the greatest risk, and analyzing how traffic patterns shifted during peak times. This section resembled conducting a "health assessment" of the intersection as it showed how the entire traffic system functions over a period, rather than solely during a handful of separate event

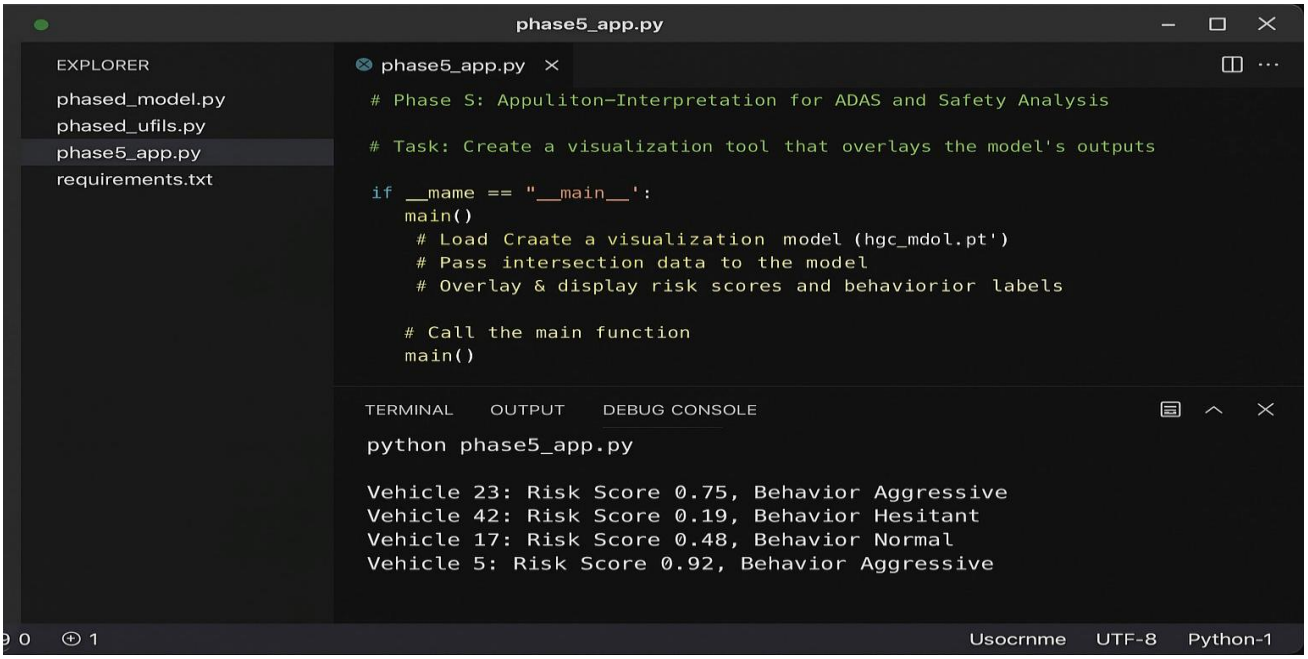






9.3 ADAS Counterfactual Simulation

The concluding section of this phase focused on examining how the dangerous events at the intersection could have altered if an ADAS feature had been implemented. After pinpointing near-miss scenarios based on the model's metrics, I conducted counterfactual simulations by altering vehicle paths to reflect common ADAS interventions. These measures involved braking sooner, less acceleration, or automated alerts that might encourage a driver to respond more quickly. Following these modifications, I re-evaluated the risk metrics to determine if the hazardous scenario would persist. In numerous instances, even minor decreases in speed or braking half a second sooner led to significant outcomes—near-misses were entirely prevented or turned into much safer encounters. These findings clearly showed the possible influence of ADAS in crowded, mixed-traffic settings. The simulations assisted in measuring how preventive technologies could diminish risk, providing a practical viewpoint on how ADAS could enhance safety at intersections similar to the one analyzed



10 . Summary

This project aims to comprehend and forecast risks at congested road intersections utilizing solely aerial video data. The process starts with charting the physical configuration of the intersection and labeling all road users to establish a solid static and dynamic basis. Leveraging this data, a tailored YOLO-based perception system identifies vehicles and pedestrians instantly, while a tracking pipeline translates their pixel movements into precise real-world paths. The trajectories are subsequently examined to calculate essential safety metrics like PET, TTC, gap acceptance, and driver compliance, aiding in the detection of near-miss incidents and dangerous behavior trends. The project's most sophisticated component employs a Heterogeneous Graph Transformer, which simulates the interactions among various agents and infrastructure to forecast short-term risks and classify driving behavior. Ultimately, the system's results are displayed on the original video, combined for intersection-level examination, and utilized to model how ADAS actions could avert possible collisions. This project establishes a comprehensive end-to-end framework—from perception to risk forecasting—demonstrating how computer vision and graph-based models can enhance intersection design and the development of next-generation ADAS systems

Conclusion

In the "Crossroads: Dynamic Intersection Risk Analysis for Proactive ADAS" project, I acquired hands-on insights into the functionality of contemporary traffic safety and ADAS technologies. I discovered how to transform standard intersection footage into valuable insights through object detection, multi-object tracking, and kinematic analysis.

I also learned how to develop a comprehensive analytical pipeline gradually, from data preparation to sophisticated graph-based prediction models. This project enhanced my abilities in data engineering, system architecture, troubleshooting, and assessing driver conduct and safety hazards.

This project enhanced my understanding of the crucial role of proactive ADAS systems in accident prevention, anticipating near-miss situations, and elevating safety at crowded intersections under real-world circumstances

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